

LESSON 8.12

FINDING SURFACE AREA AND VOLUME



LESSON INTRODUCTION

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula.

LEARNING TARGETS

- I can find the surface area of right rectangular prisms and right rectangular pyramids.
- I can find the volume of right rectangular prisms.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.

WORKING TOWARD

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

ESSENTIAL QUESTIONS

- What is different between surface area and volume?
- How do you find the volume and surface area of a rectangular prism?
- How do you find the surface area of a rectangular pyramid?

LESSON NARRATIVE

This lesson reviews skills and concepts from this unit including finding volume and surface area of rectangular prisms and pyramids and finding the missing dimension when given the volume. Through collaboration and guided practice, students strengthen previously learned skills including drawing nets and finding missing dimensions, surface area, lateral area, and volume of rectangular pyramids and rectangular prisms, including within real-world contexts.

COMPONENT SUMMARY

8.12.1 WARM-UP (~5 MIN.)

Would You Rather explanation

8.12.2 COLLABORATION (10 MIN.)

Students collaborate to solve volume and surface area problems

8.12.3 GUIDED PRACTICE (10 MIN.)

Students create nets to find surface area and volume of prisms and pyramids

8.12.4 YOUR TURN! (15 MIN.)

Individual practice finding surface area, volume, and missing dimensions

8.12.5 WRAP-UP (~5 MIN.)

Self-reflection on understanding of learning targets and how to overcome a barrier to learning

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Lateral face
- Surface area
- Volume

MATERIALS

- Four-function calculator

ADDITIONAL PREPARATION

- Real-world examples to use for comparing volume and surface area may be beneficial.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may attempt to multiply the given volume when prompted to find the missing dimension (rather than dividing or identifying the value of the missing multiple).
- Students may get stuck explaining how they would overcome learning barriers from the lesson.

THINGS TO CONSIDER

- Students may need a refresher on how to find missing dimensions when given volume for 8.12.2 Collaboration.
- Consider having students practice the various vocabulary terms related to three-dimensional figures and finding surface area and volume to foster independence and problem solving. Provide models of prisms and pyramids, then prompt students to recognize the length, width, height, faces, surface area, and volume before 8.12.2 Collaboration.
- In some of the real-world contexts the student will not be required to find the total surface area. Students may get confused with the wording “surface area” versus “surface area that should be painted”. Help students understand that in the real-world it is not always necessary to find the area of all of the surfaces.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

Consider using a Compare and Connect routine to stimulate mathematical discussion about surface area and volume. During 8.12.2 Collaboration and 8.12.3 Guided Practice, use the practice problems to have students compare these two mathematical concepts and to connect them to each other through real-world examples and through the lens of prisms versus pyramids.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

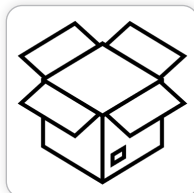
MA.K12.MTR.7.1 Real-World Contexts

8.12.4 Your Turn! engages students in real-world examples on finding surface area and volume with prisms and pyramids. This section highlights math through real-world examples, as students explore how the skills they have learned in this unit translate to real-world scenarios.

DIFFERENTIATE INSTRUCTION

Consider providing a table or similar strategy for graphically organizing information to assist students as they solve the problems for 8.12.2 Collaboration. The table supports students to visually organize given information, partner input, and student calculations throughout the problem-solving process.

	<i>L</i>	<i>W</i>	<i>H</i>	<i>V</i>	<i>SA</i>
<i>Box 1</i>					



8.12.1 WARM-UP: WOULD YOU RATHER?

Component Overview: Students explain why they would rather choose a cake with the indicated dimensions and slices to share. Consider providing students time to connect the real-world scenario to area and compare reasoning through discussion.

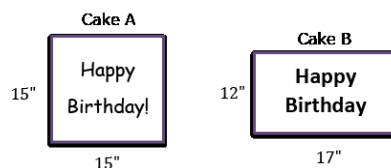
DIFFERENTIATE INSTRUCTION

As an alternate method to elicit student responses, post an image of each cake on opposite sides of the room and have the students sit on the side of the room according to which cake they would rather have. Allow them to post their answers and explanations in one location, such as on chart paper, or engage in a class discussion to provide students opportunity to explain their reasoning verbally.



8.12.1 Warm-Up: Would You Rather?

1. Cake A and Cake B are shown below. Cake A is cut into 8 equal slices. Cake B is cut into 6 equal slices. Would you rather have a slice from Cake A or Cake B?

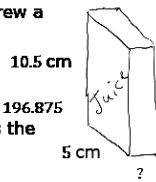


Answers will vary.
I would choose Cake A because it is the biggest.
I would choose Cake B because the slices are bigger.



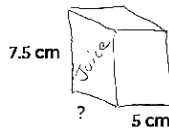
8.12.2 Collaboration: Which Juice Box Is Better?

Julius is designing a juice box. He drew a model of his box.



1. If the juice box has a volume of 196.875 cubic centimeters (cm^3), what is the width of the box?
3.75 cm
2. How much packaging board would Julius need to make a juice box with these dimensions, not taking into consideration any overlap of the faces?
221.25 cm^2

Julius wondered how the juice box would be affected if he changed the dimensions, but kept the volume the same.



3. What is the new length?
5.25 cm
4. How much packaging board would Julius need to make the new juice box?
206.25 cm^2
5. If Julius wants to use the least amount of material to construct the juice box, which juice box should he choose? Explain your reasoning.
He should use the second juice box because the surface area is smaller.

8.12.2 COLLABORATION: WHICH JUICE BOX IS BETTER?

Component Overview: Students collaborate to find a missing dimension, then the volume and surface area of two real-world examples of prisms: juice boxes. Have students work with partners to compare the juice boxes, including justifying which prism will use the least amount of material to construct. Consider having students use a table or chart to organize needed information and found measurements. Students can optionally create a poster comparing the two juice boxes, including the organized table of measurements and sketches of the nets/prisms and their work for questions 2 and 4 and reasoning for question 5. Then the posters can be placed around the classroom for a gallery walk.

TEACHER MOVES

Compare and Connect

Prior to beginning the Collaboration, prompt students to consider how volume and surface area are similar and different. Students may point out that volume involves an amount on the inside of three-dimensional figures and that surface area involves an amount on the outside of three-dimensional figures. Consider providing real-life examples of volume, such as the amount of juice inside of a juice box, and surface area, such as the cardboard that creates the juice box.

DIFFERENTIATE INSTRUCTION

!!! Consider providing a table or graphic organizer, such as the table below, for students to use as they problem-solve on question 5. This strategy can be used with any problem in this lesson as an added visual support.

8.12.3 GUIDED PRACTICE: SURFACE AREA AND VOLUME

Component Overview: Students work in partners to practice finding surface area and volume with prisms and pyramids, including identifying the figure to draw a labeled net and needed dimensions. Consider providing chart paper and/or graph paper as students draw the nets and label the measurements.

ENRICHMENT

As partnerships begin working on these problems, pose these questions to foster strategic math reasoning and various approaches to finding surface area and volume of different figures:

- What are the differences between rectangular pyramids and rectangular prisms?
- How do the differences between pyramids and prisms affect how you find the volume? The surface area?
- What different strategies do you know to find the surface area/volume of a prism/pyramid? Which strategy makes the most sense for this problem? Why?

DIFFERENTIATE INSTRUCTION

Consider providing students with chart paper to complete the tables and draw the nets for questions 1 and 2.

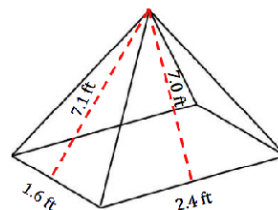
Having ample space to draw a net is important as students solidify the visual representation of a three-dimensional figure as a collection of two-dimensional shapes. Seeing both figures from questions 1 and 2 on the chart paper can help strengthen conceptual understanding as students compare and connect the figures and approaches to finding surface area and volume.



8.12.3 Guided Practice: Surface Area and Volume

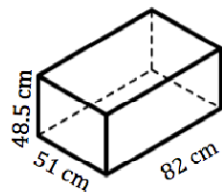
With your partner, complete the chart for the three-dimensional figures below.

1.



What is the type of figure?	Pyramid
Draw a net or the individual faces of the figure. Label the measurements.	
Find the surface area of the figure.	32 ft ²

2.



What is the type of figure?	Rectangular Prism
Draw a net of the figure or the individual faces. Label the measurements.	
Find the surface area of the figure.	21,265 cm ²
Find the volume of the figure.	$V = 51 \times 82 \times 48.5 = 202,827 \text{ cm}^3$

8.12.3 GUIDED PRACTICE: SURFACE AREA AND VOLUME (CONTINUED)

QUESTIONING

Prompt students to find an example of a rectangular prism or pyramid in the classroom and complete a table like question 2 as additional practice.

8.12.4 YOUR TURN!

Component Overview: Students individually find volume and surface area by applying their understanding to four real-world problems related to rectangular prisms and pyramids. As students work on these problems, consider running a small group to provide additional support to students who may still have difficulty finding missing dimensions, creating nets, and/or finding surface area and volume.

DIFFERENTIATE INSTRUCTION

For question 1, it may be helpful to first have visual students draw a net to represent the pyramid. This practice may help students recognize how the lateral triangular faces have two different heights and how many faces there will be with those measurements.

TEACHER MOVES

For question 1b, students without background knowledge on skyscrapers may not recognize which sections of the skyscraper will be covered in glass. Connect mathematics to the real-world by showing a picture of a pyramid-shaped skyscraper and asking:

- What do you notice about the picture? Which parts have glass?
- Which parts of our model will not have glass? How do you know?
- How can you use this knowledge to find the amount of surface area that will be covered in glass?

DIFFERENTIATE INSTRUCTION

Consider bringing a cooler to class for students to analyze and manipulate. Though the dimensions between the model and actual cooler will be different, students can visualize how the inside and outside of the cooler have different dimensions because of the thickness of the sides, which may aid problem-solving for question 2.

TEACHER MOVES

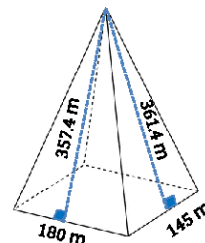
Compare and Connect

For question 2c, students may notice that the question is not just asking for the volume of the inside of the cooler, but it is also asking for the difference between the two volumes. As students work individually or during small group discussions, students can recognize different approaches to comparing the volumes, including finding the volume of the inside of the cooler and subtracting it from the volume of the outside of the cooler.



8.12.4 Your Turn!

1. Nikolas is constructing a skyscraper with walls made of only glass that is 350 meters (m) high and is in the shape of a rectangular pyramid.



- a. What is the surface area of the skyscraper?
 $142,835 \text{ m}^2$
- b. Using the model shown, determine how much of the surface area will be made of glass.
 $116,735 \text{ m}^2$

2. A cooler has a length of 78 centimeters (cm), a height of 74 cm, and a width of 40 cm.



- a. Find the surface area of the cooler.
 $23,704 \text{ cm}^2$
- b. Find the volume of the cooler.
 $V = 78 \times 40 \times 74 = 230,880 \text{ cm}^3$
- c. If the inside of the cooler has a length of 62 cm, a height of 58 cm, and a width of 28 cm, how much smaller is the volume of the inside of the cooler than the volume of the entire cooler?



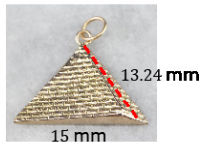
$$\begin{aligned}
 V_{\text{inside}} &= 62 \times 28 \times 58 = 100,688 \text{ cm}^3 \\
 V_{\text{entire}} - V_{\text{inside}} &= 230,880 \text{ cm}^3 - 100,688 \text{ cm}^3 \\
 &= 130,192 \text{ cm}^3
 \end{aligned}$$

3. Tamyra is creating a gardening box that needs to have a volume of 13.5 ft^3 .
- a. If the box has a width of 3 feet and a length of 2 feet, what is the height?



$h = 2.25 \text{ ft}$

- b. Tamyra decided to paint the exterior walls of the gardening box and needs to know how much paint to buy. What is the surface area of the area she wants to paint?
- 22.5 ft^2
4. Ryan makes custom jewelry and recently received an order for a pendant in the shape of a rectangular pyramid. Ryan designed a hollow gold pendant to have a square base with a side length of 15 millimeters (mm) and a slant height of 13.24 mm.



How much gold does Ryan need to make the pendant?

622.2 mm^2

8.12.4 YOUR TURN! (CONTINUED)

QUESTIONING

As an optional hands-on extension, prompt students to recreate a life-size model of one of the prisms or pyramids from questions 2, 3, or 4. This kinesthetic practice will not only further understanding in creating accurate nets, but also in measurement, as students need to use a ruler to measure precisely. Students can use cardstock paper, tape, modeling clay, or other classroom models to build their representations.



8.12.5 Wrap-Up: Reflection

1. Rate yourself on each of the concepts.

- 4 – I can do this on my own.
 3 – I can do this but need more practice.
 2 – I need help getting started.
 1 – I do not understand yet.

Concept	Rating			
I can find the surface area of a right rectangular pyramid using a net or the figure.	1	2	3	4
I can find the surface area of a right rectangular prism using a net or the figure.	1	2	3	4
I can find the volume of a right rectangular prism.	1	2	3	4

2. A barrier to my learning today was ...
Answers will vary.
Trying to remember how to find the area of a triangle.
Trying to remember how many triangles a pyramid has.
3. I will try to overcome this by ...
Answers will vary.
Remembering that the area of a triangle is half of a rectangle.
Remembering the number of triangles on a pyramid is determined by the rectangular base which has 4 sides.

8.12.5 WRAP-UP: REFLECTION

Component Overview: Students rate their conceptual understanding of the lesson's learning targets using a number scale, then describe a barrier to learning from today's lesson and a way they will try to overcome it.

DIFFERENTIATE INSTRUCTION

For students who get stuck on question 3, provide ideas for ways students can overcome barriers to learning by writing a list on the whiteboard of shared ideas from the class.

Examples:

- Texting with my friend about question 2 when I get home
- Watching the Study Expert videos after dinner to try to learn from them
- Attending tutoring on _____ (day)